07/04/2025

CODE-A

Aakash

Medical | IIT-JEE | Foundations

***Corporate Office*: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015**

AIM – 720

(*Advanced* INTENSIVE *Mastery for* 720)

MM : 720

Redemption Test

Time : 180 Mins.

Answers

1.	(3)	37.	(1)	73.	(3)	109.	(3)	145.	(3)
2.	(4)	38.	(1)	74.	(3)	110.	(3)	146.	(3)
3.	(4)	39.	(1)	75.	(1)	111.	(1)	147.	(4)
4.	(3)	40.	(2)	76.	(2)	112.	(1)	148.	(2)
5.	(4)	41.	(3)	77.	(2)	113.	(1)	149.	(3)
6.	(4)	42.	(1)	78.	(3)	114.	(3)	150.	(2)
7.	(2)	43.	(4)	79.	(2)	115.	(1)	151.	(2)
8.	(3)	44.	(4)	80.	(2)	116.	(3)	152.	(3)
9.	(1)	45.	(2)	81.	(3)	117.	(2)	153.	(1)
10.	(4)	46.	(1)	82.	(3)	118.	(4)	154.	(3)
11.	(4)	47.	(4)	83.	(2)	119.	(4)	155.	(4)
12.	(2)	48.	(3)	84.	(2)	120.	(4)	156.	(1)
13.	(4)	49.	(2)	85.	(1)	121.	(3)	157.	(1)
14.	(3)	50.	(3)	86.	(2)	122.	(3)	158.	(3)
15.	(4)	51.	(3)	87.	(1)	123.	(2)	159.	(4)
16.	(2)	52.	(3)	88.	(4)	124.	(3)	160.	(2)
17.	(4)	53.	(4)	89.	(4)	125.	(2)	161.	(1)
18.	(4)	54.	(4)	90.	(4)	126.	(2)	162.	(1)
19.	(1)	55.	(4)	91.	(2)	127.	(4)	163.	(2)
20.	(3)	56.	(1)	92.	(3)	128.	(1)	164.	(2)
21.	(4)	57.	(2)	93.	(3)	129.	(4)	165.	(4)
22.	(3)	58.	(4)	94.	(3)	130.	(3)	166.	(2)
23.	(4)	59.	(4)	95.	(4)	131.	(3)	167.	(1)
24.	(4)	60.	(2)	96.	(3)	132.	(3)	168.	(3)
25.	(3)	61.	(1)	97.	(4)	133.	(3)	169.	(2)
26.	(3)	62.	(1)	98.	(3)	134.	(1)	170.	(3)
27.	(1)	63.	(2)	99.	(1)	135.	(3)	171.	(2)
28.	(3)	64.	(1)	100.	(3)	136.	(3)	172.	(3)
29.	(3)	65.	(2)	101.	(1)	137.	(4)	173.	(4)
30.	(1)	66.	(2)	102.	(2)	138.	(4)	174.	(3)
31.	(3)	67.	(2)	103.	(1)	139.	(3)	175.	(3)
32.	(4)	68.	(3)	104.	(3)	140.	(2)	176.	(3)
33.	(1)	69.	(2)	105.	(3)	141.	(2)	177.	(4)
34.	(1)	70.	(4)	106.	(1)	142.	(3)	178.	(3)
35.	(3)	71.	(2)	107.	(2)	143.	(4)	179.	(3)
36.	(4)	72.	(3)	108.	(2)	144.	(2)	180.	(3)

Hints and Solutions

PHYSICS

(1) **Answer :** (3)
Solution:
$\overrightarrow{r} = -(2\hat{i} + 2\hat{j} + \hat{k})$
$r = \sqrt{4+4+1} = 3$
$\overrightarrow{F} = \frac{-q_1 q_2}{4\pi\varepsilon_0 3^3}(2\hat{i} + 2\hat{j} + \hat{k})$
$\overrightarrow{F} = \frac{-q_1 q_2}{108\pi\varepsilon_0}(2\hat{i} + 2\hat{j} + \hat{k})$

(2) **Answer :** (4)
Solution:
Use $W = q\Delta V$
Sol.: $V_A = V_B = V_C = V_E = V_D$
$W = q \times 0 = 0$

(3) **Answer :** (4)
Solution:
$V = \frac{kq}{r} + \frac{-kq}{r} + \frac{kq}{r} - \frac{kq}{r} = 0$

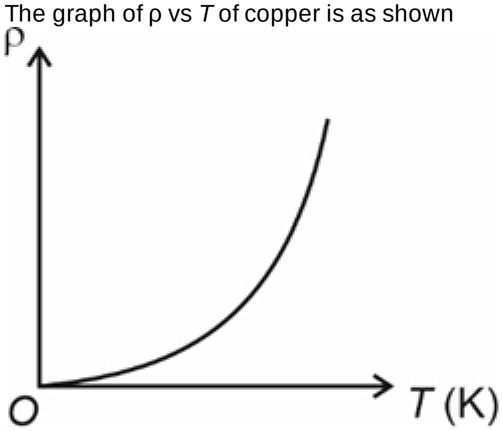
(4) **Answer :** (3)
Solution:
When separation between the plates is halved, capacitance becomes two times. So charge stored in capacitor is doubled.
$\frac{F}{F_1} = \frac{q^2}{4q^2}$
$F_1 = 4F$

(5) **Answer :** (4)
Solution:

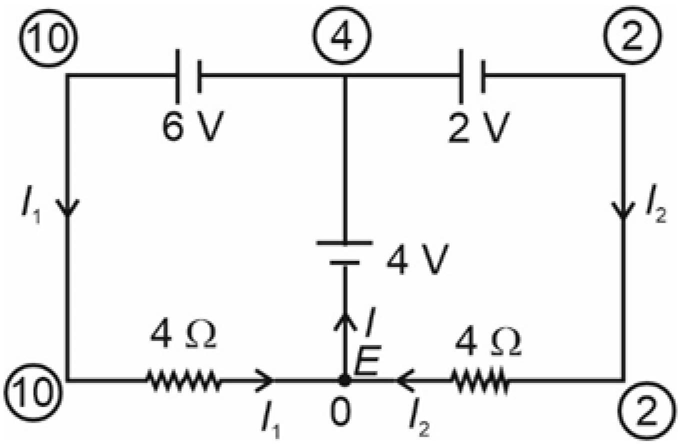
To determine ε, the information about angle between $\overrightarrow{v}$ and $\overrightarrow{B}$ along with angle between $\overrightarrow{v} \times \overrightarrow{B}$ and $\overrightarrow{l}$ is needed.

(6) **Answer :** (4)
Solution:
Since, μ_r is always positive
∴ μ is also always positive.

(7) **Answer :** (2)
Solution:
The graph of ρ vs T of copper is as shown



(8) **Answer :** (3)
Solution:
Potential of various points in the circuit (modified) is encircled.



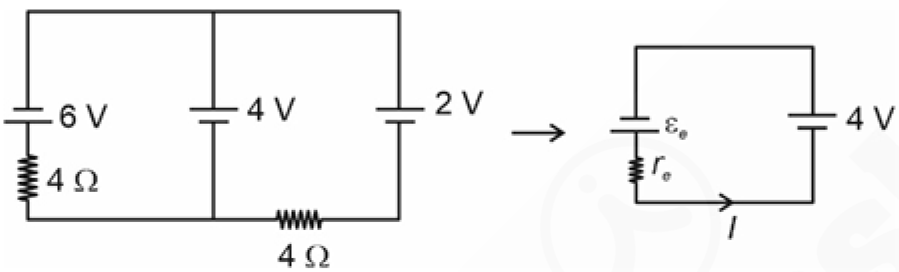
Here, $I = I_1 + I_2$

$I = \frac{10-0}{4} + \frac{2-0}{4}$

$I = 2.5 + 0.5$

$I = 3$ A

Method-2



$\varepsilon_e = \frac{\frac{6}{4} - \frac{2}{4}}{\frac{1}{4} + \frac{1}{4}} = 2$ V

$r_e = 2\ \Omega$

$I = \frac{2+4}{2}$ A $= 3$ A

(9) **Answer :** (1)

Solution:

We know, $I = \frac{V}{R} = \frac{1000}{100} = 10$ A

$\therefore B = 4\pi \times 10^{-7} \times 10^{3} \times 10$

$B = 4\pi \times 10^{-3}$ T

(10) **Answer :** (4)

Solution:

Statement I is wrong because V_{rms} = 220 V in domestic AC circuit and peak voltage = $220\sqrt{2}$ V.

There could be a half time period in AC current which may have zero average value.

(11) **Answer :** (4)

Solution:

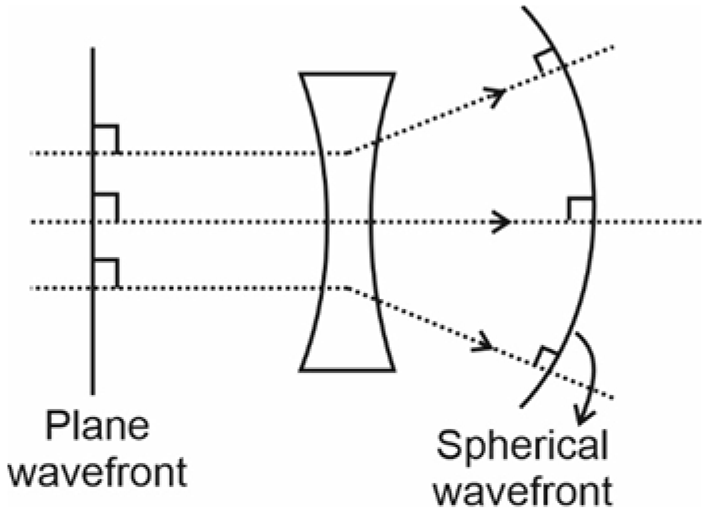
Ray does not suffer any deviation on entering the lens. $\therefore \mu_1 = \mu_2$.

The ray leaving second surface is bending towards the normal.

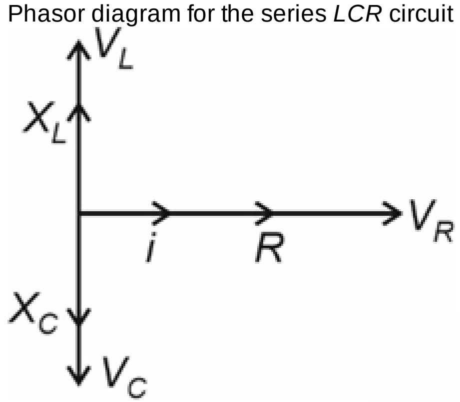
$\therefore \mu_3 > \mu_2$ (as μ_3 is denser than μ_2)

(12) **Answer :** (2)

Solution:

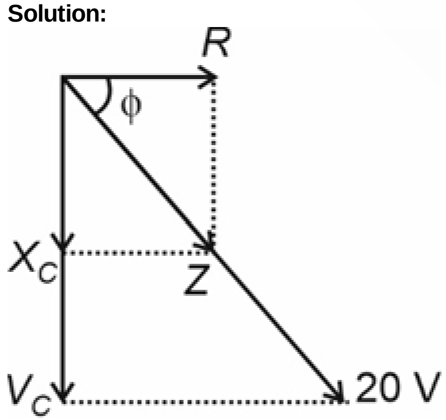


(13) **Answer :** (4)
Solution:
Phasor diagram for the series LCR circuit is



All statements are true.
Current through C and R are in same phase because current is same through each elements.

(14) **Answer :** (3)
Solution:



$\tan\phi = \frac{X_C}{R} = \frac{3}{4} = \tan 37^{\circ}$ ⇒ $\varphi = 37^{\circ}$
∴ $90^{\circ} - \varphi = 53^{\circ}$
V_C lags E_{max} by 53°
$V_C = 20\sin\phi = 20 \times \frac{3}{5} = 12\ \text{V}$
Voltage across capacitor = $V_C \sin(\omega t - 53^{\circ})$
= $12\sin(\omega t - 53^{\circ})$ V

(15) **Answer :** (4)
Solution:
Initial momentum, $p = \frac{E}{c}$

Change in momentum, $\Delta p = 0 - \left(-\frac{E}{c}\right) = \frac{E}{c}$

Force exerted, $F = \frac{\Delta p}{\Delta t} = \frac{E}{ct}$

Average pressure exerted = $\frac{F}{A} = \frac{E}{ctA}$

(16) Answer : (2)
Solution:
When contact angle is less than 90°, then shape of meniscus is concave upwards.

(17) Answer : (4)
Solution:
By Einstein's equation
$(K.E.)_{max} = hf - \varphi$
∴ By quadrupling the frequency, (K.E.) becomes more than 4 times.

(18) Answer : (4)
Solution:
Total binding energy (B.E.) = $\Delta m(u) \times 930$ MeV
= 0.077 × 930 MeV
∴ Binding energy per nucleon = $\frac{0.077 \times 930}{11}$
= $7 \times 10^{-3} \times 930$
= 6510×10^{-3}
= 6.51 MeV

(19) Answer : (1)
Solution:
$\frac{1}{G} = \frac{1}{S} - \frac{1}{R}$
$\Rightarrow \frac{1}{S} = \frac{1}{R} + \frac{1}{G}$
$\Rightarrow \frac{1}{S} = \frac{1}{600} + \frac{1}{3} = \frac{201}{600}$
$\Rightarrow S = \frac{600}{201}\ \Omega$

(20) Answer : (3)
Solution:
Nuclear forces are independent of charge and are type of non-central forces.

(21) Answer : (4)
Solution:
The current under reverse bias is voltage independent upto breakdown voltage (V_{br}).
Even a slight increase in bias voltage at $V = V_{br}$, the diode reverse current increases sharply.

(22) Answer : (3)
Solution:
$E = -13.6 \times \frac{Z^2}{n^2}$ eV
Kinetic energy in 1st orbit of H atom = 13.6 eV = K
Potential energy in 1st orbit of He^+ atom $= -\frac{13.6 \times 2^2}{1^2} \times 2$ eV
∴ PE = – (13.6 eV) × 8 = – 8K

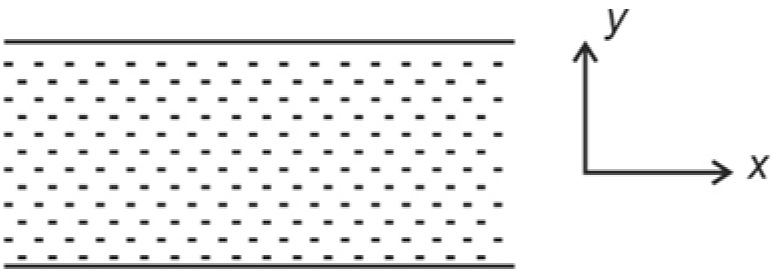
(23) Answer : (4)
Solution:
$\tan 90^\circ = \frac{Q \sin\theta}{P + Q\cos\theta}$ or $P + Q\cos\theta = 0$
$\Rightarrow \cos\theta = \frac{-P}{Q}$
$R = \frac{Q}{2} = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$
$\frac{Q^2}{4} = P^2 + Q^2 + 2PQ \times \frac{-P}{Q}$
On solving
$\frac{P}{Q} = \frac{\sqrt{3}}{2}$
$\cos\theta = \frac{-\sqrt{3}}{2} = \cos 150^\circ$

θ = 150°

(24) **Answer :** (4)
Solution:
$\overrightarrow{V}_{m/w} = 3\hat{j}$

$\overrightarrow{V}_{b/w} = -4\hat{i}$

$\overrightarrow{V}_{m/b} = 3\hat{j} - (-4\hat{i}) \ = 3\hat{j} + 4\hat{i}$



$\left|\overrightarrow{V}_{m/b}\right| = \sqrt{9+16}$ = 5 km/h

Direction $\theta = \tan^{-1}\left(\frac{3}{4}\right)$

θ = 37°, with the river flow.

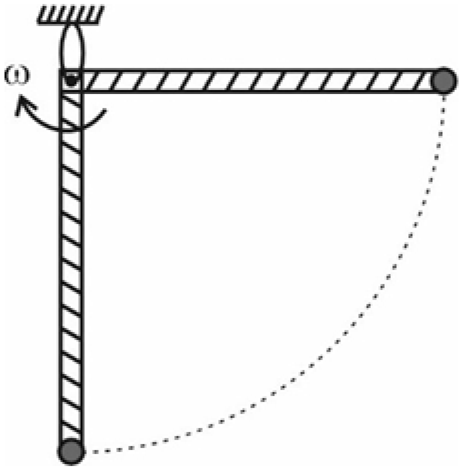
(25) **Answer :** (3)
Solution:
$\frac{\text{joule ohm}}{\text{second × volt}} = \frac{\text{joule}}{\text{second × ampere}} = \frac{\text{joule}}{\text{coulomb}} = \text{volt}$

(26) **Answer :** (3)
Solution:
Dimensions of a physical quantity are the powers to which base quantities are raised to express that quantity. For example
$[F] = [MLT^{-2}]$
According to principle of homogeneity
If $x + y = z$
Then $[x] = [y] = [z]$

(27) **Answer :** (1)
Solution:
At origin x-coordinate = 0
Sol.: $x = \sin t - \sqrt{3}\ \cos t$
$\sin t = \sqrt{3}\ \cos t$
$\tan t = \sqrt{3}$
$t = \frac{\pi}{3}$ s

(28) **Answer :** (3)
Solution:
$I_1\omega_1 = I_2\omega_2$
$MR^2\omega = \left(MR^2 + \frac{1}{2}\frac{M}{2}R^2\right)\omega'$
$= \frac{5MR^2}{4}\omega'$
$\therefore\ \omega' = \frac{4}{5}\omega$

(29) **Answer :** (3)
Solution:
$KE_i + PE_i = KE_f + PE_f$



⇒ $0+0=\frac{1}{2}I\omega^2+\left(-2\frac{mgL}{2}-mgL\right)$

⇒ $\frac{1}{2}I\omega^2=2mgL$

⇒ $\frac{1}{2}\left[\frac{2ML^2}{3}+ML^2\right]\omega^2=2mgL$

⇒ $\frac{5}{6}mL^2\omega^2=2mgL$

⇒ $\omega=\sqrt{\frac{12}{5}\frac{g}{L}}$

(30) **Answer :** (1)
Solution:
Tension in the wire will be maximum at the lowest point while it will be minimum at the highest point. Hence, the wire will be least likely to break at highest point.

(31) **Answer :** (3)
Solution:
Energy radiated $E = P \times t$
$= 20 \times 10^3 \times 3600 \times 3$
$= 216 \times 10^6$ J
$= 2.16 \times 10^8$ J
$= 1.35 \times 10^{27}$ eV

(32) **Answer :** (4)
Solution:
For solid sphere, $I'_{tangent} = \frac{2}{5}MR^2+MR^2$
$=\frac{7}{5}MR^2=M\,k_1^2$
For hollow sphere, $I''_{tangent} = \frac{2}{3}MR^2+MR^2$
$=\frac{5}{3}MR^2=M\,k_2^2$
$\frac{Mk_1^2}{Mk_2^2}=\frac{\left(\frac{7}{5}MR^2\right)}{\left(\frac{5}{3}MR^2\right)}=\frac{7\times3}{5\times5}=\frac{21}{25}$
$\frac{k_1}{k_2}=\sqrt{\frac{21}{25}}$

(33) **Answer :** (1)
Solution:
$84 - T = 4a$ …(i)
$T - 30 = 5a$ …(ii)
On solving ⇒ $54 = 9a$
$a = 6$ m s^{-2}
$T = 60$ N

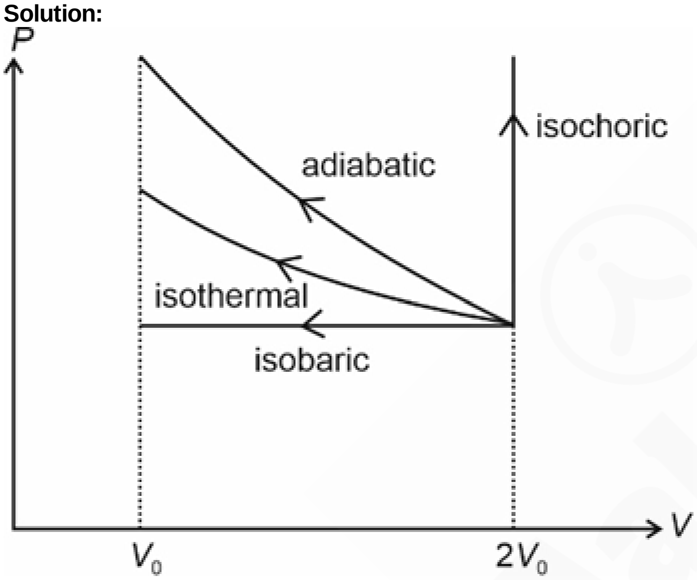
(34) **Answer :** (1)
Solution:
Needle floats due to the force of surface tension.

(35) **Answer :** (3)
Solution:
Force required to just pull it out = 2 × S × 4R = 8SR.

(36) **Answer :** (4)
Solution:
Shear stress = $\frac{F\sin\theta}{A}$

(37) **Answer :** (1)
Solution:
$h=\frac{2S}{R\rho g}$ ⇒ The liquid does not overflow because $hR = h'R'$ = constant.
i.e., liquid rises to full length of tube and radius of curvature becomes larger.

(38) **Answer :** (1)
Solution:
$\frac{1}{2}m\,V_e^2-\frac{GMm}{R}=0\Rightarrow \text{P.E.}=\frac{1}{2}m\,V_e^2=\frac{1}{2}\times2\times100\times100$
= – 10 kJ

(39) **Answer :** (1)
Solution:


Area under adiabatic process is maximum.

(40) **Answer :** (2)
Solution:
When both S.H.M.s are having same period and along the same path, then resultant amplitude is given by
$R=\sqrt{A_1^2+A_2^2+2A_1A_2\cos\phi}$
Sol.: $y_1=4\sin(\omega t)$ and $y_2=-3\sin(\omega t)$
⇒ φ = 180°
∴ $R = A_1 - A_2$ = 4 – 3 = 1 m

(41) **Answer :** (3)
Solution:
Both springs are in parallel combination, $k_{eq} = k_1 + k_2 = k + k = 2k$
Angular frequency of SHM,
$\omega=\sqrt{\frac{k_{eq}}{m}}=\sqrt{\frac{2k}{m}}=\sqrt{\frac{2\times10}{5}}=2\ \text{s}^{-1}$
Amplitude, A = 5 m
∴ $v_{max} = \omega A$ = 2 × 5 = 10 m/s

(42) **Answer :** (1)
Solution:
A diathermic wall allows flow of heat but not matter across it.

(43) **Answer :** (4)

Solution:
A combination of two SHM is periodic only if freqquencies of both are same.

(44) **Answer :** (4)
Solution:
Wave motion in option (1), is longitudinal.
Wave motion in option (2), is both transverse and longitudinal.
Wave motion in option (3), is both transverse and longitudinal.

(45) **Answer :** (2)
Solution:
Statement I is wrong as no real gas is ideal at room temperature.
Statement II is wrong as at low pressures and high temperatures, the molecules are far apart and molecular interactions are negligible.

CHEMISTRY

(46) **Answer :** (1)
Hint:
$2C_2H_6 + 7O_2 \rightarrow 4\,CO_2 + 6H_2O$
Solution:
Mole of C_2H_6 (ethane) = $\frac{180}{30} = 6$
Mole of O_2 required = $\frac{7}{2} \times 6 = 21$
Volume of O_2 required at STP
= 21 × 22.4 = 470.4 L

(47) **Answer :** (4)
Hint:
Number of atoms = Moles × N_A × atomicity
Solution:
7 g of nitrogen gas = $\frac{7}{28} \times N_A \times 2$
⇒ 0.5 N_A
34 g of ammonia gas = $\frac{34}{17} \times N_A \times 4$
= 8 N_A
1.6 g of oxygen gas = $\frac{1.6}{32} \times N_A \times 2$
⇒ 0.1 N_A
2 g of hydrogen gas = $\frac{2}{2} \times N_A \times 2$
= 2 N_A

(48) **Answer :** (3)
Hint:
Bohr's theory could not explain ability of atoms to form molecules by chemical bonds
Solution:
Bohr's theory was unable to explain the splitting of spectral lines in the presence of magnetic field or an electric field.

(49) **Answer :** (2)
Hint:
Energy of photon = hν
Solution:
Power of bulb = 100 Js^{-1}
Energy of one photon = $E = \frac{hc}{\lambda}$
$= \frac{6.626 \times 10^{-34} \times 3 \times 10^{8}}{6.626 \times 10^{-12}} = \frac{10^{-34} \times 3 \times 10^{8}}{10^{-9}}$
= 3 × 10^{-17} J
Number of photons emitted = $\frac{100}{3 \times 10^{-17}}$
= 33.33 × 10^{17}
= 3.33 × $10^{18} s^{-1}$

(50) Answer : (3)

Hint:

Ionization enthalpy generally increases across the period.

Solution:

$1s^2 2s^2 2p^3$ represents electronic configuration of N and it has stable half-filled configuration of *p*-orbital so it has maximum ionization enthalpy.

(51) Answer : (3)

Hint:

Cerium (Ce) is *f*-block element

Solution:

• *s* and *p* block elements are called representative elements. Bismuth (Bi) and rubidium (Rb) are *p* and *s* block elements respectively.

• Ge and Sb are semi-metals or metalloids.

• Ti is *d* block element.

(52) Answer : (3)

Hint:

Higher charge increases the energy and decreases stability of the species.

Solution:

Formal charge helps in the selection of lowest energy structure from a number of possible Lewis structures. Generally the lowest energy structure is the one with the smallest formal charges on atoms.

(53) Answer : (4)

Hint:

Orbital dipole due to lone pair has to be considered in both NH_3 and NF_3.

Solution:

In case of NH_3 the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the N–H bonds, whereas in NF_3 the orbital dipole is in the direction opposite to the resultant dipole moment of three N–F bonds.

(54) Answer : (4)

Hint:

The change in entropy of the reversible system can be explained using, $\Delta \mathrm{S} = \frac{\mathrm{q_{rev}}}{\mathrm{T}}$.

Solution:

For ideal gas; undergoing reversible change dU = 0 and $\Delta S_{total} = 0$.

For ideal gas undergoing irreversible change,

dU = 0; and $\Delta S_{total} \neq 0$.

(55) Answer : (4)

Hint:

Heat added to the system can influence molecular motions and hence randomness.

Solution:

As entropy is inversely proportional to the temperature $\Delta \mathrm{S} = \frac{\mathrm{q_{rev}}}{\mathrm{T}}$, so, at low temperature, there is more randomness caused when heat is added to system.

(56) Answer : (1)

Hint:

Ion from weak electrolyte undergoes hydrolysis.

Solution:

$NH_4Cl(aq)$ is acidic due to cationic hydrolysis.

(57) Answer : (2)

Hint:

$PbCl_2$ is sparingly soluble salt; so common ion can influence its solubility.

Solution:

$\mathrm{PbCl_2} \rightleftharpoons \mathrm{Pb^{2+}} + 2\,\mathrm{Cl^-}$

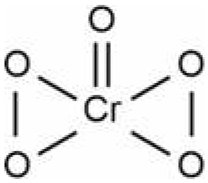
$MgCl_2$ has 0.04 M of Cl^-, hence cause the maximum suppression of $PbCl_2$. Hence, least solubility.

(58) Answer : (4)

Hint:

Structure of CrO_5 has two peroxy linkage

Solution:

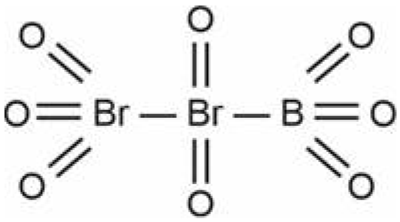


Cr is in +6 oxidation state.

(59) **Answer :** (4)

Hint:

Tribromooctaoxide is



Solution:

Oxidation state of terminal Br = +6

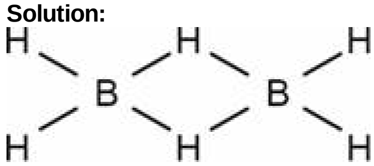
Oxidation state of middle Br = +4

(60) **Answer :** (2)

Hint:

Diborane has 3c-2e bonds due to electron deficient nature.

Solution:



It has two 3-centre-2-electron bonds and four 2-centre-2-electron bonds.

(61) **Answer :** (1)

Solution:

Order of melting point for group 13 is

B > Al > Tl > In > Ga

(62) **Answer :** (1)

Hint:

Vapours of the liquid with higher boiling point condense before the vapours of the liquid with lower boiling point.

Solution:

The vapours rising up in the fractionating column become richer in more volatile component.

(63) **Answer :** (2)

Solution:

Paper chromatography is a type of partition chromatography.

(64) **Answer :** (1)

Hint:

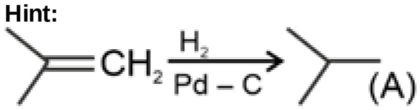
More symmetrical is the compound, better is the crystal packing and higher is the melting point.

Solution:

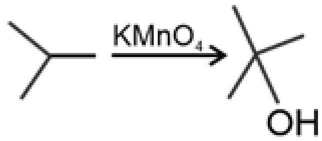
Compound	**Melting point (K)**
Heptane	182.4
Hexane	178.5
2, 2- Dimethylpropane	256.4
2-Methylbutane	113.1

(65) **Answer :** (2)

Hint:



Solution:



(66) **Answer :** (2)
Hint:
For ideal solutions, intermolecular attractive forces between A-A and B-B are nearly, equal to those between A-B.
Solution:
For ideal solution;
$\Delta_{mix}H = 0$, $\Delta_{mix}V = 0$
$\Delta_{mix}G < 0$, $\Delta_{mix}S > 0$

(67) **Answer :** (2)
Hint:
24 g of methanol is present in 100 g of solution.
Solution:
Mass of H_2O = 100 – 24 = 76 g

$$\text{Molality} = \frac{\text{Mole of methanol} \times 1000}{\text{Mass of water (in g)}}$$

$$= \frac{24 \times 1000}{32 \times 76}$$

Molality = 9.87 m

(68) **Answer :** (3)
Hint:
If E°_{cell} is positive the cell reaction will be spontaneous and disproportionation reaction will take place
Solution:
- $\overset{+1}{I}O^- \rightarrow \overset{+5}{I}O_3^-$; E° = –0.14 V

  $\overset{+1}{I}O^- \rightarrow I_2^0$; E° = 0.45 V

  $E^\circ_{cell} = (0.45 - 0.14)\ V = 0.31$ V
- $\frac{1}{2}I_2^0 \rightarrow \overset{+1}{I}O^-$; E° = –0.45 V
- $\frac{1}{2}I_2^0 \rightarrow I^-$; E° = 0.54 V

$E^\circ_{cell} = 0.54 - 0.45 = 0.09$ V

Both IO^- and I_2 will undergo disproportionation reaction.

(69) **Answer :** (2)
Hint:
Conductivity of CuO is 1×10^{-7} S m^{-1}
Solution:

Material	Conductivity (S m^{-1})
Copper	5.9×10^3
Iron	1.0×10^3
CuO	1×10^{-7}

(70) **Answer :** (4)
Solution:
$SF_4 \rightarrow sp^3d$, see-saw
$NH_4^+ \rightarrow sp^3$, Tetrahedral
$BrF_3 \rightarrow sp^3d$, bent T-shaped
$XeF_4 \rightarrow sp^3d^2$, Square planar

(71) **Answer :** (2)
Hint:

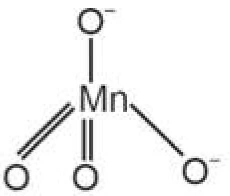
For a bimolecular elementary reaction
A + B → Products
Rate of the reaction can be expressed as,
$\text{Rate} = PZ_{AB}\, e^{-E_a/RT}$
Solution:
Z_{AB} = Collision frequency of reactants, A and B
P = Probability or steric factor
$e^{-E_a/RT}$ = Fraction of molecules with energies equal to or greater than E_a.

(72) **Answer :** (3)
Hint:
The plot of ln k v/s $\frac{1}{T}$ gives a straight line
$\ln k = \frac{-E_a}{RT} + \ln A$
Slope = $\frac{-E_a}{R}$ and intercept = ln A
Solution:
$\ln k = \ln A - \frac{E_a}{RT}$
Slope = $\frac{-E_a}{R}$
$\frac{-E_a}{R} = -4 \times 10^2$
$E_a = 4 \times 10^2 \times 8.314$
E_a = 3.325 kJ mol^{-1}

(73) **Answer :** (3)
Solution:

- $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$
- $2NaNO_2 + 2FeSO_4 + 3H_2SO_4 \rightarrow$
  $Fe_2(SO_4)_3 + 2NaHSO_4 + 2H_2O + 2NO$
- $2Pb(NO_3)_2 \xrightarrow{673K} 4NO_2 + 2PbO + O_2$
- $4HNO_3 + P_4O_{10} \rightarrow 4HPO_3 + 2N_2O_5$

(74) **Answer :** (3)
Solution:
Sulphur dioxide gas reacts with chlorine in the presence of charcoal (which acts as a catalyst) to give sulphuryl chloride, SO_2Cl_2.

(75) **Answer :** (1)
Hint:
The manganate ion is MnO_4^{2-}, in which central Mn atom is sp^3 hybridised.
Solution:

Tetrahedral manganate (green) ion

(76) **Answer :** (2)
Hint:
$KMnO_4$ on heating at 513 K gives K_2MnO_4, MnO_2 and O_2.
Solution:
$2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

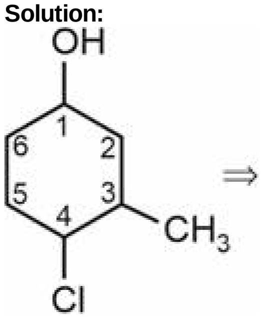
(77) **Answer :** (2)

Hint:
As field strength of ligand increases CFSE increases.
Solution:
CFSE increases with chelation, synergic bonding and oxidation state of central metal ion.
Hence order of CFSE
$[Fe(CN)_6]^{3-} > [Fe(CN)_6]^{4-} > [Fe(C_2O_4)_3]^{3-} > [Fe(H_2O)_6]^{3+}$

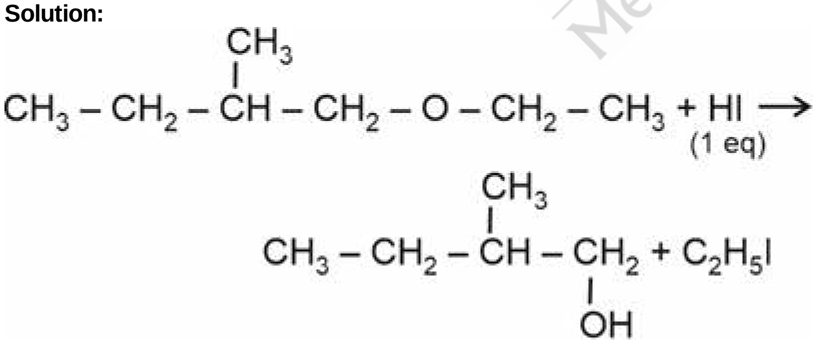
(78) **Answer :** (3)
Hint:
Polydentated ligands form chelate ring.
Solution:
Hydrazine $H_2N - NH_2$ is a monodentate ligand.

(79) **Answer :** (2)
Solution:
Weak bases are good leaving groups. Order of leaving group tendency is
$F^{\ominus} > \overset{\ominus}{O}H > \overset{\ominus}{N}H_2 > \overset{\ominus}{C}H_3$

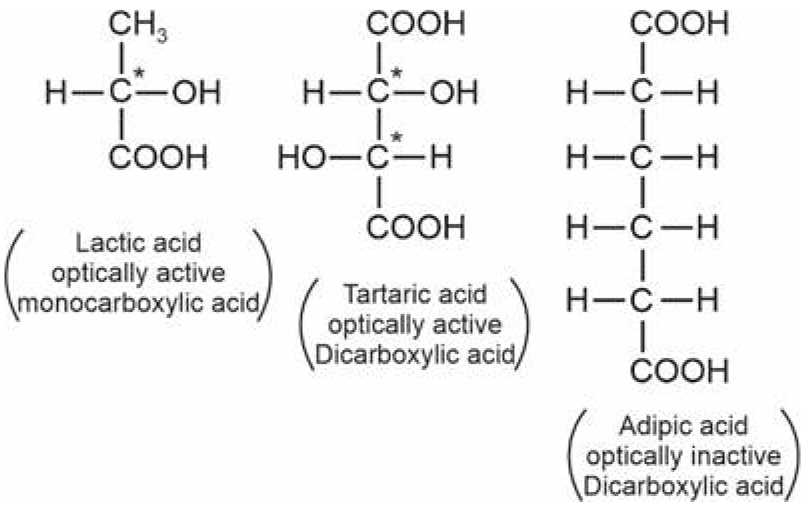
(80) **Answer :** (2)
Solution:
Selectivity ratio for monochlorination with respect to 1º, 2º, 3º carbon is 1 : 3.8 : 5 respectively.
% primary halide = $\frac{9\times1}{9+5}\times100 = 64.28\%$
% tertiary halide = $\frac{5}{14}\times100 = 35.72\%$

(81) **Answer :** (3)
Hint:
IUPAC naming is done on the basis to lowest locant rule.
Solution:


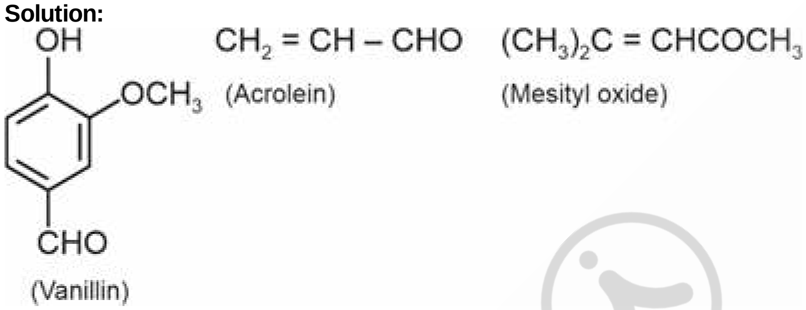
4-chloro-3-methylcyclo hexane-1-ol

(82) **Answer :** (3)
Hint:
Reaction of ether with HI is nucleophilic substitution in which O – R bond breaks.
Solution:


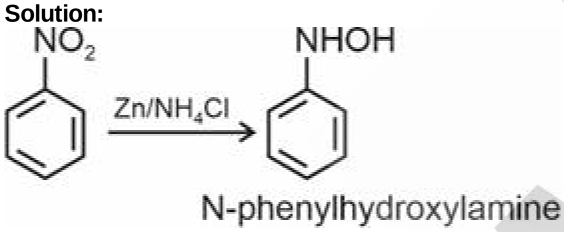
(83) **Answer :** (2)
Hint:
Tartaric acid and adipic acid are dicarboxylic acids.
Solution:



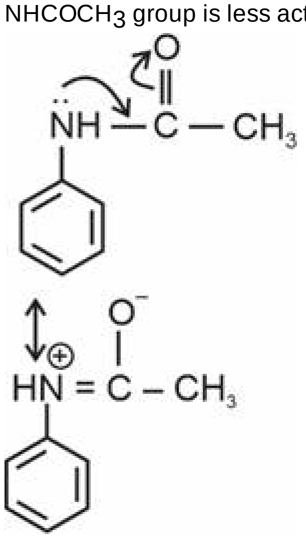
(84) **Answer :** (2)
Solution:



(85) **Answer :** (1)
Solution:



(86) **Answer :** (2)
Solution:
$NHCOCH_3$ group is less activating than amino group.



Lone pair on Nitrogen is in conjugation with carbonyl group in acetylated aniline.

(87) **Answer :** (1)
Hint:

Pb^{2+} reacts with K_2CrO_4 to form yellow precipitate of lead chromate.

Solution:

$Pb^{2+} + K_2CrO_4 \rightarrow PbCrO_4 + 2K^+$

Lead chromate is soluble in hot sodium hydroxide solution.

$\mathrm{PbCrO_4 + 4NaOH \rightleftharpoons Na_2[Pb(OH)_4] + Na_2CrO_4}$

(88) **Answer :** (4)

Hint:

Ceric ammonium nitrate reacts with alcohol to form red coloured complex.

$(NH_4)_2[Ce(NO_3)_6] + 3ROH \rightarrow [Ce(NO_3)_4(ROH)_3] + 2NH_4NO_3$

Solution:

Phenols form violet coloured compound with freshly prepared $FeCl_3$ solution.

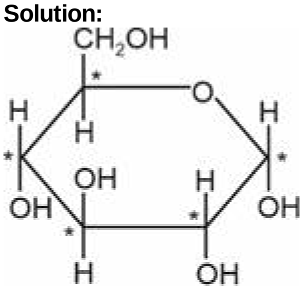
$6C_6H_5OH + FeCl_3 \rightarrow [Fe(C_6H_5O)_6]^{3-} + 3HCl + 3H^+$

(89) **Answer :** (4)

Hint:

In the glucopyranose structure, 5 carbon atoms and one oxygen atom are present in the ring.

Solution:



α-D(+) - Glucopyranose

Five carbon atoms marked (*) are chiral

(90) **Answer :** (4)

Hint:

A unit formed by attachment of a base to 1′ position of sugar is known as nucleoside.

Solution:

When nucleoside is linked to phosphoric acid at 5′ position of sugar moiety, a nucleotide is formed.

BIOLOGY

(91) **Answer :** (2)

Solution:

Outer layer of seed coat is testa.

(92) **Answer :** (3)

Solution:

The MMC undergoes meiosis and forms a linear tetrad of four haploid megaspores. Out of which, one remains functional (chalazal end) and three degenerate (micropylar end).

(93) **Answer :** (3)

Solution:

In embryo sac, the egg apparatus is 3-celled and 3-nucleate.

(94) **Answer :** (3)

Solution:

Methanogens are bacteria of marshy habitats, capable of converting CO_2, methanol and formic acid into methane. These bacteria are obligate anaerobes.

(95) **Answer :** (4)

Solution:

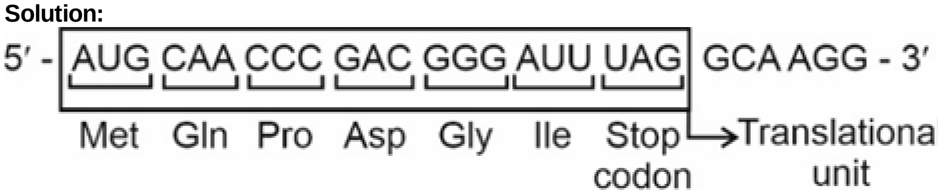
Viruses being non-cellular agents were not included in the five-kingdom classification.

(96) **Answer :** (3)

Solution:

Rhizopus belongs to zygomycetes in which mycelium is coenocytic, motile zoospores are absent, zygospore is the site of meiosis.

(97) Answer : (4)
Solution:



(98) Answer : (3)
Solution:
Binding of mRNA with smaller subunit of ribosome is energy independent process.

(99) Answer : (1)
Solution:
During splicing, introns are removed by small nuclear RNA (snRNA) and protein complex called small nuclear ribonucleoproteins or SnRNPs (Snurps).

(100) Answer : (3)
Solution:
Sugar-phosphate backbone is present in both RNA and DNA. The plane of one base pair stacks over the other in double helix. This in addition to H-bonds, confers stability to the helical structure. Presence of thymine at the place of uracil also confers additional stability to DNA.

(101) Answer : (1)
Solution:
5386 nucleotides are present in φ × 174 bacteriophage.

(102) Answer : (2)
Solution:
(i) *Aspergillus niger* is used for the production of citric acid and gluconic acid.
(ii) Amylases are produced by *Aspergillus*, *Rhizopus* and *Bacillus* species.
(iii) Acetic acid is produced by *Acetobacter aceti.*

(103) Answer : (1)
Solution:
Toddy is a traditional drink in some parts of southern India. It is prepared by fermentation of the sap from palm (*Caryota urens*) trees.

(104) Answer : (3)
Solution:
Dachigam National Park is home to musk deer or hangul.

(105) Answer : (3)
Solution:
The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.

(106) Answer : (1)
Solution:
Cranial meninges consists of an outer layer called dura mater, a very thin middle layer called arachnoid and an inner layer called pia mater.

(107) Answer : (2)
Solution:
Three major regions make up the brain stem, *i.e.*, midbrain, pons and medulla oblongata. Cerebellum has very convoluted surface.
Two cerebral hemispheres are connected by a tract of nerve fibres called corpus callosum. Amygdala, hippocampus and hypothalamus are parts of limbic lobe in forebrain.

(108) Answer : (2)
Solution:
The movements that result in a change of place or location is termed as locomotion. Activities like search of food, shelter, mate, suitable breeding grounds, favourable climatic conditions, involve both locomotion and movement.

(109) Answer : (3)
Solution:

Skeletal muscles fibres are multi-nucleated in nature.
Stomach, intestine and urinary bladder are visceral organs and possess smooth muscles.

(110) Answer : (3)
Solution:
Flagellar movement helps in the swimming of spermatozoa, maintenance of water current in the canal system of sponges and in locomotion of protozoans like *Euglena*.

(111) Answer : (1)
Solution:
MSH affects pigmentation of skin. Parathyroid hormone increases the Ca^{2+} levels in the blood. PTH acts on bones and stimulates the process of bone resorption. Thyrocalcitonin stimulates Ca^{2+} deposition in bones. Hence, to treat the fracture, thyrocalcitonin should be injected to the patient.

(112) Answer : (1)
Solution:
Pituitary gland is located in a bony cavity called sella tursica. Pineal gland is located on the dorsal side of forebrain.
Hypothalamus is the basal part of diencephalon.
Thyroid gland is located on either side of the trachea.

(113) Answer : (1)
Solution:
Conditional reabsorption of Na^+ and water takes place in DCT. Maximum reabsorption of filtrate takes place in PCT.
Collecting duct does not form the part of renal tubule.

(114) Answer : (3)
Solution:
Inspiration is initiated by the contraction of diaphragm which increases the volume of thoracic chamber in antero-posterior axis. The contraction of external inter-costal muscles lifts up the ribs and sternum causing an increase in volume of the thoracic chamber in the dorso-ventral axis.

(115) Answer : (1)
Solution:
Trachea, primary, secondary and tertiary bronchi and initial bronchioles are supported by incomplete cartilaginous rings.
Pharynx, terminal bronchioles and external nostrils do not have C-shaped cartilaginous rings.

(116) Answer : (3)
Solution:
In an aquatic ecosystem, the pyramid of biomass may be inverted. Biomass of zooplanktons is higher than that of phytoplanktons as life span of former is longer and the latter multiply much faster though having shorter life span.

(117) Answer : (2)
Solution:

Grasses →	Grasshopper →	Birds →	Hawk.
T_1	T_2	T_3	T_4
40,000 J	4000 J	400 J	40 J
40 kJ	4 kJ	0.4 kJ	0.04 kJ

(118) Answer : (4)
Solution:
Quasi-fluid nature of lipids, enables lateral movement of protein within the overall bilayer and this ability to move within the membrane is measured as fluidity.

(119) Answer : (4)
Solution:
Microbodies are present in both plant and animal cells. They lack DNA.

(120) Answer : (4)
Solution:
Trans face of Golgi complex gives rise to the secretory vesicles.

(121) Answer : (3)
Solution:
Number of meiotic divisions required to make x number of seeds
= $x+\frac{x}{4}=100+\frac{100}{4}=125$

(122) Answer : (3)

Solution:
During interkinesis, there is no replication of DNA.

(123) Answer : (2)
Solution:
M-phase is the most dramatic period of the cell cycle which involves a major reorganisation of virtually all components of the cell.

(124) Answer : (3)
Solution:
Because of the specific requirements and the need of water for fertilization, the spread of living pteridophytes is limited and restricted to narrow geographical regions.

(125) Answer : (2)
Solution:
Chlorella is a unicellular alga.

(126) Answer : (2)
Solution:
Bulliform cells are present on adaxial epidermal cells.

(127) Answer : (4)
Solution:
Dicot root has small inconspicuous pith. Dicot stem has starch filled endodermis. In monocot roots, xylem bundles are usually more than six (polyarch).

(128) Answer : (1)
Solution:
Casparian strips are seen in the endodermal layer of roots.

(129) Answer : (4)
Solution:
The descending order of substrate w.r.t. their RQ value

Malic acid	>	Glucose	>	Proteins	>	Tripalmitin
1.33		1		0.9		0.7

(130) Answer : (3)
Solution:

PEPcase	–	Present in mesophyll cell of C_4 plant
ATP synthase	–	Embedded in the thylakoid membrane
Assimilatory power	–	ATP and $NADPH_2$
Carotenoids	–	Prevent photo-oxidation of chlorophyll a.

(131) Answer : (3)
Solution:
Thrombocytes or platelets are cell fragments produced from megakaryocytes (special cells in the bone marrow). Only leucocytes are nucleated cells among all the formed elements.

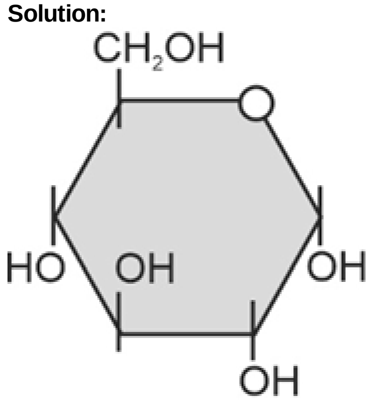
(132) Answer : (3)
Solution:
Each artery and vein consists of three layers; an inner lining of squamous endothelium, the tunica intima; a middle layer of smooth muscle and elastic fibres, the tunica media; and an external layer of fibrous connective tissue with collagen fibres, the tunica externa. The tunica media is comparatively thin in the veins.

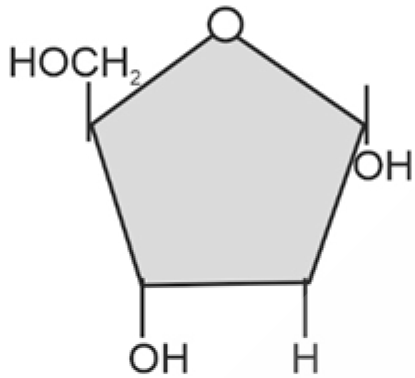
(133) Answer : (3)
Solution:
In systemic circulation, blood flows from:- Left ventricle → Aorta → Body tissues → Vena cava → Right atrium.
In pulmonary circulation, blood flows from:-
Right ventricle → Pulmonary artery → Lungs → Pulmonary vein → Left atrium.

(134) Answer : (1)
Solution:
Aquatic amphibians are ammonotelic and many terrestrial amphibians are ureotelic. In many bony fishes, ammonia diffuses out through the gill surfaces.

(135) Answer : (3)
Solution:
Stretch receptors present on the walls of the urinary bladder send signals to the CNS. The CNS passes on motor signals to initiate contraction of smooth muscles of urinary bladder and simultaneous relaxation of urethral sphincter causing the release of urine.

(136) Answer : (3)
Solution:
A general rule of thumb is that rate doubles or decreases by half for every 10°C change in temperature in either direction.
In the mentioned case, temperature is decreased by 10°C, therefore, the rate of reaction decreases by half (2X becomes X).

(137) Answer : (4)
Solution:


(Glucose)


(2′deoxyribose)

The sugar found in polynucleotides is either ribose (a monosaccharide pentose) or 2′ deoxyribose. A nucleic acid containing deoxyribose is called deoxyribonucleic acid (DNA) while that which contains ribose is called ribonucleic acid (RNA).

(138) Answer : (4)
Solution:
In ‘peroxidase’ and ‘catalase’, which catalyze the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of the active site of the enzyme.
Prosthetic groups are organic compounds and are distinguished from other co-factors in that they are tightly bound to the apoenzyme.

(139) Answer : (3)
Solution:
Substituted purines : Adenine and guanine
Substituted pyrimidines : Cytosine and thymine
Given, guanine = 35%
According to Chargaff’s rule:-
[Guanine] = [Cytosine]
[Adenine] = [Thymine]
Therefore, G and C would be 35% each and A and T would be 15% each.
Thus, cytosine and thymine would be 35 + 15 = 50.

(140) Answer : (2)
Solution:

Component	**% of the total cellular mass**
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

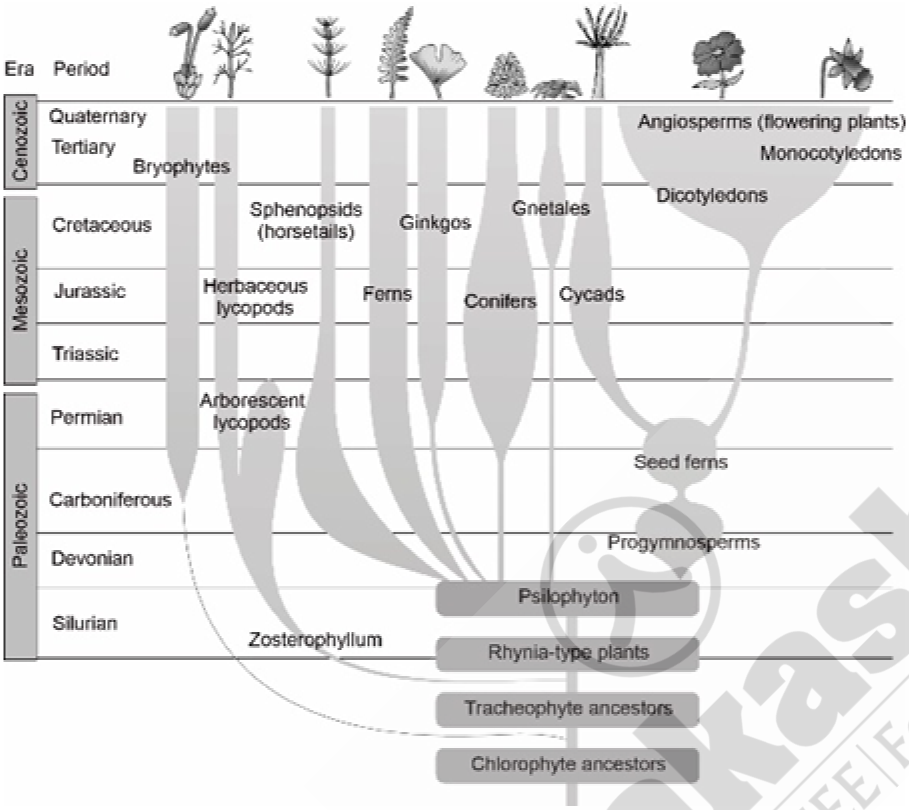
Collagen – Protein, Glycogen – Carbohydrate, Palmitic acid-Lipid, DNA-Nucleic acid, Calcium-Ion.

(141) Answer : (2)
Solution:

In all connective tissues, except blood, the cells secrete fibres of structural proteins called collagen or elastin. The fibres provide strength, elasticity and flexibility to the tissues.

(142) Answer : (3)
Solution:
Repeated activation of the muscles can lead to accumulation of lactic acid due to anaerobic breakdown of glycogen in them, causing fatigue.

(143) Answer : (4)
Solution:



(144) Answer : (2)
Solution:
In 1953, S. L. Miller, an American scientist, created conditions prevailing in the primitive Earth in a laboratory scale. He created electric discharge in a closed flask containing CH_4, H_2, NH_3 and water vapour at 800°C. He observed the formation of amino acids.

(145) Answer : (3)
Solution:
Given that 25% of the population exhibits the recessive phenotype (aa).
Thus, $q^2 = 0.25$
$q = \sqrt{0.25} = 0.5$
Thus, the frequency of the recessive allele (q) = 0.5.

(146) Answer : (3)
Solution:
IUDs do not inhibit oogenesis.
Administration of progestogens or progestogen-estrogen combinations or IUDs within 72 hours of coitus have been found to be effective as emergency contraceptives as they could be used to avoid possible pregnancy due to rape or casual unprotected intercourse.
Contraceptives are not regular requirements for the maintenance of reproductive health.

(147) Answer : (4)
Solution:
Saheli – the new oral contraceptive for the females contain a non-steroidal preparation. It is ‘once a week’ pill with very few side effects and high contraceptive value. It does not inhibit ovulation. Contraceptive injections contain progestogens alone or combinations of estrogen and progestogens.

(148) Answer : (2)

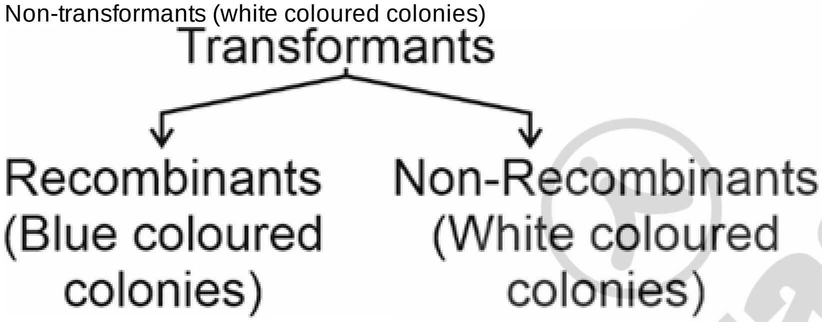
Solution:
In humans, embryo enters the uterus in morula stage but implantation takes place in blastocyst stage within the endometrium of uterus. Trophoblast gets attached within the endometrium of uterus and inner cell mass differentiates into the ectoderm and endoderm.

(149) Answer : (3)
Solution:
In adult humanmale, primary and secondary sex organs are located in the pelvis region. Scrotum helps in maintaining temperature lower than the normal internal body temperature, necessary for spermatogenesis.

(150) Answer : (2)
Solution:
Uterus, cervix and vagina are unpaired structures whereas ovary, oviduct and mammary glands are paired structures in reproductive system of human females.

(151) Answer : (2)
Solution:
Penis is made up of special tissue that helps in its erection to facilitate insemination in vagina. Seminal plasma is rich in fructose, calcium and certain enzymes.

(152) Answer : (3)
Solution:
The Lac-Z gene codes for the β-galactosidase. The enzyme utilises its substrate to produce blue coloured product.
In the given case,
Non-transformants (white coloured colonies)



(153) Answer : (1)
Solution:
Gene gun or biolistics, electroporation and micro-injection are the direct methods of gene transfer.
Retroviruses have been disarmed and used to deliver desirable genes into animal cells.

(154) Answer : (3)
Solution:
*Eco*RI is the first restriction enzyme to be isolated from RY 13 strain of *Escherichia coli*.
*Hin*d II was isolated from strain Rd of *Haemophilus influenzae*.
*Eco*RI recognises the following palindromic nucleotide sequence and cuts in between G and A and produces sticky ends.
5’ G↓AATTC 3'
3’ CTTAA↑G 5’

(155) Answer : (4)
Solution:
The cells can be multiplied in continuous culture system wherein the used medium is drained from one side while fresh medium is added from the other to maintain the cells in their physiologically most active log/exponential phase.
After completion of the biosynthetic phase, the product has to be subjected through a series of processes before it is ready for marketing as a finished product.
The processes include separation and purification, which are collectively termed as downstream processing.
The downstream processing and quality control testing vary from product to product.

(156) Answer : (1)
Solution:
Oxidation of each molecule of Acetyl-CoA in Krebs cycle produces one molecule of GTP, three molecules of $NADH^+ + H^+$ and one $FADH_2$ molecule.

(157) Answer : (1)
Solution:
The total gain of ATP in the glycolysis is 4 molecules.

(158) Answer : (3)

Solution:
Nutrients (macro and micro essential elements) are required by plants for the synthesis of protoplasm and act as source of energy. Every plant has optimum temperature range best suited for plant growth.

(159) Answer : (4)
Solution:
ABA inhibits the protein and RNA synthesis.

(160) Answer : (2)
Solution:
In Solanaceae, berry fruits and axile placentation are found.

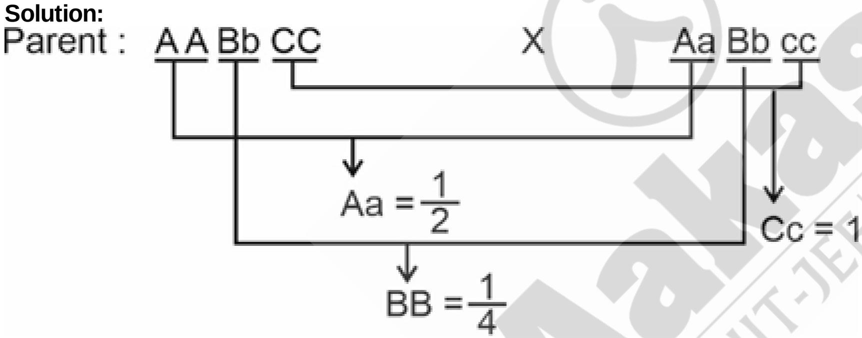
(161) Answer : (1)
Solution:
In Marigold, basal placentation is found in ovary. The placenta develops at the base of the ovary.

(162) Answer : (1)
Solution:
Zygomorphic flowers can be divided into two similar halves only in one particular plane. e.g., Pea, Gulmohur, Bean and *Cassia*.

(163) Answer : (2)
Solution:
In Mango and coconut, the fruits develop from monocarpellary ovary and are one seeded. In Mango, the pericarp is well differentiated into outer thin epicarp, middle fleshy edible mesocarp and an inner stony endocarp.

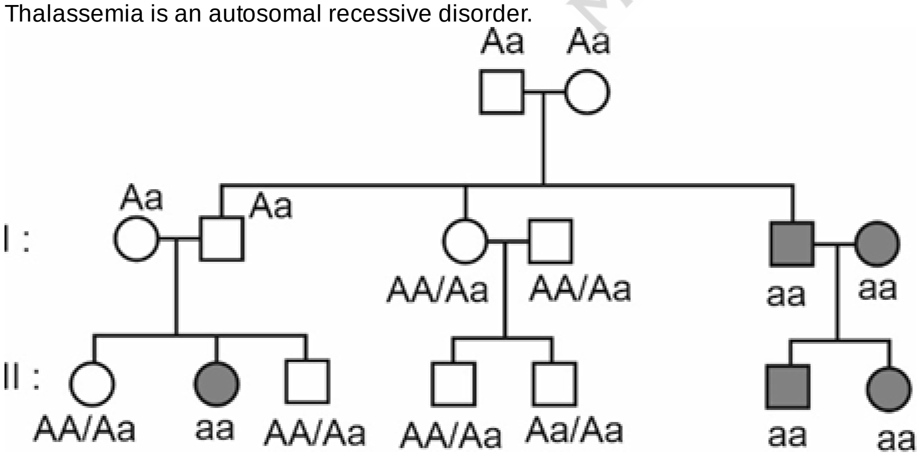
(164) Answer : (2)
Solution:
People affected with haemophilia-B lacks plasma thromboplastin. Cystic fibrosis is an autosomal recessive disorder.

(165) Answer : (4)
Solution:



Probability of having AaBBCc
$= \frac{1}{2} \times \frac{1}{4} \times 1 = \frac{1}{8}$

(166) Answer : (2)
Solution:
Thalassemia is an autosomal recessive disorder.



(167) Answer : (1)
Solution:
45 with X0 is the chromosomal complement of Turner's syndrome.

(168) Answer : (3)
Solution:
Arrangement of taxonomic categories in descending order in animal is
Kingdom, Phylum, Class, Order, Family, Genus, Species.

(169) Answer : (2)
Solution:
Intraspecific competition is more acute as compared to interspecific competition.

(170) Answer : (3)
Solution:
Relative density is a good measure to find total density of fish in a lake by counting the number of fishes caught per trap.

(171) Answer : (2)
Solution:
Restriction enzymes cleave the DNA fragments at their respective recognition sequences.
The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution.
After precipitation using chilled ethanol, the DNA can be seen as a collection of fine threads in the suspension and can be spooled up and twisted around a glass rod. This step is called spooling.
Transformation is the procedure through which a piece of DNA is introduced into a host bacterium.

(172) Answer : (3)
Solution:
Transgenic rats, rabbits, pigs, sheep, cows and fishes have been produced, although over 95% of all existing transgenic animals are mice.
Transgenic mice are being used to test the safety of polio vaccine.

(173) Answer : (4)
Solution:
Bioremediation is the use of microorganisms to remove pollutants. Biofortification is the method of breeding crops to increase their nutritional value.
At present, about 30 recombinant therapeutics have been approved for human-use, the world over.
Regeneration of whole plant from explant is the part of tissue culture.

(174) Answer : (3)
Solution:
Acquired immunity is pathogen specific. It is characterised by memory. This means when our body encounters a pathogen for the first time, it produces a response called the primary response which is of low intensity. Subsequent encounter with the same pathogen elicits a highly intensified secondary or anamnestic response. This is ascribed to the fact that our body appears to have memory of the first encounter.

(175) Answer : (3)
Solution:
AIDS is caused by the Human Immuno Deficiency Virus (HIV), a member of a group of viruses called retroviruses, which have an envelope enclosing the RNA genome.
HIV contains two identical molecules of single-stranded RNA and two molecules of reverse transcriptase.
HIV evades the immune system by constantly undergoing mutation.

(176) Answer : (3)
Solution:
Health increases longevity of people and reduces infant and maternal mortality.
The innate defences of our body like skin, mucous membranes, anti-microbial substances present in our tears, saliva and the phagocytic cells help to block the entry of pathogens into our body.
Health does not simply mean ‘absence of diseases’ or ‘physical fitness’. It could be defined as a state of complete physical, mental and social well-being.
The B-lymphocytes produce an army of proteins in response to pathogens in our blood to fight with there. T-Cell themselves do not secrete antibodies but help B-cells to secrete there.

(177) Answer : (4)
Solution:
In cockroach, oesophagus opens into a sac-like structure called crop, used for storing of food. The crop is followed by gizzard or proventriculus. Gizzard helps in grinding the food particles. The entire foregut is lined by cuticle. A ring of 6-8 blind tubules called hepatic or gastric caeca is present at the junction of foregut and midgut, which secrete digestive juice. At the junction of midgut and hindgut, is present another ring of 100-150 yellow coloured thin filamentous Malpighian tubules. They help in removal of excretory products from haemolymph.

(178) Answer : (3)
Solution:

An open type of blood circulatory system is exhibited by arthropods, hemichordates, non -cephalopod molluscs in which the blood is pumped out of the heart and the cells and tissues are directly bathed in it.
Vertebrates and cephalochordates exhibit closed type of blood circulation in which the blood is circulated through a series of vessels of varying diameters (veins, arteries and capillaries).

(179) Answer : (3)
Solution:

Bony fishes	**Cartilaginous fishes**
Mouth is mostly terminal	Mouth is located ventrally
Skin is covered with cycloid/ctenoid scales	Skin is tough, containing minute placoid scales
Fertilisation is usually external	Fertilisation is usually internal
Claspers are absent	Pelvic fins of male bear claspers

(180) Answer : (3)
Solution:
In frogs, the brain is divided into forebrain, midbrain and hindbrain. Forebrain has olfactory lobes, paired cerebral hemispheres and unpaired diencephalon. The midbrain is characterised by a pair of optic lobes. Hindbrain consists of cerebellum and medulla oblongata.
Eyes and internal ears are well-organised structures whereas sensory papillae, taste buds and nasal epithelium are cellular aggregations around nerve endings.